

AI in Healthcare

Evidence, workflow, and human judgment

Patient intake

Diagnostics

Operations

Monitoring

Agenda

I Healthcare workflows: intake, diagnostics, operations, monitoring

II AI applications: imaging, prediction, multimodal GenAI

III Main case: pediatric chest X-ray prioritization

IV Notebook demonstration: technical evidence in five stages

V Applied AI Studio: evidence becomes workflow action

VI Human authority, privacy, regulation, and monitoring

SECTION I • 15 min

Healthcare is four connected workflows

Patient intake, diagnostics, operations, and monitoring. AI enters a task inside one of them; it does not replace the system.

Four workflow families organize the healthcare system

1 · Patient intake

Identity, consent, symptoms, routing, urgency.

2 · Diagnostics

Order, acquire, interpret, combine evidence.

3 · Operations

Scheduling, queues, capacity, and handoffs.

4 · Monitoring

Track patients, workflow, models, and safety.

A correct prediction can still fail in the workflow

It can arrive too late, reach the wrong person, or create too many alerts.

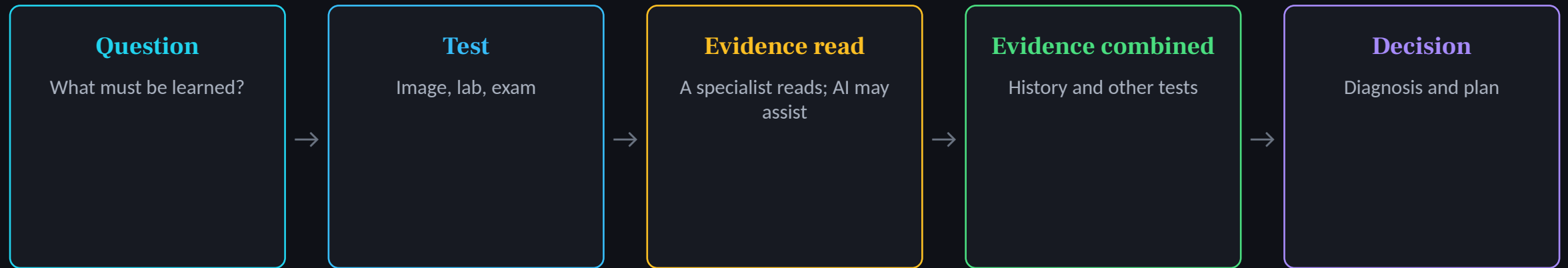
Patient intake: structure the request without losing urgency

INTAKE TASK	BEST FIRST TOOL	HUMAN BOUNDARY
Required field present?	Written rule	The requirement is explicit.
Extract referral facts	Document AI	Verify uncertainty and conflicts.
Route to specialty	Classification	Escalate ambiguous cases.
Is the patient unstable?	Clinical judgment	Keep urgent assessment human.

AI cannot fill in missing patient information

Missing symptoms, a wrong identity, or unavailable history remain missing or wrong.

Diagnostics is an evidence chain, not a single prediction



Operations determines whether useful evidence arrives in time

- 1 Queues**
Which case should be handled next?
- 2 Capacity**
Which person, room, or device is available?
- 3 Handoffs**
What evidence and responsibility move?
- 4 Exceptions**
What happens when the normal path fails?

Our classroom model changes queue order only

Radiologists still interpret each image, and clinicians still diagnose and treat.

Monitoring means more than watching model metrics

Model

AUC, sensitivity, calibration, drift, and image quality.

Workflow

Queue volume, time to read, overrides, and alerts acted on.

Patient

Time to interpretation, access, outcomes, and subgroup differences.

AUC (area under the ROC curve) is a ranking metric

It summarizes how often pneumonia-labeled images receive higher scores than normal-labeled images across all cutoffs. AUC can stay steady while queues grow, alerts go unread, or patients wait longer.

One patient crosses all four workflows



Your turn: mark the human checkpoint

A referral becomes an X-ray, a score, and a treatment decision. Work in pairs.

1 Draw four workflow families

Place intake, diagnostics, operations, monitoring.

2 Place one AI task

Name its input and output.

3 Mark two human decisions

Who interprets? Who treats?

4 Add one failure route

Unreadable image or unavailable model.

SECTION II • 18 min

Three application patterns, many narrow tasks

Medical imaging analysis, predictive analytics, and multimodal GenAI. Name the output before naming the technology.

Medical imaging AI can produce four different outputs

Classify

Label or score the whole image; it does not locate a finding.

Detect

Locate a finding with a box or point.

Segment

Outline pixels belonging to a structure.

Prioritize

Use a score to reorder work.

Predictive analytics estimates what may happen next

QUESTION	TIME HORIZON	ACTION IT COULD SUPPORT
Will this patient deteriorate?	Hours	Escalate observation
Will this patient be readmitted?	30 days	Plan follow-up
Will this appointment be missed?	Days	Target outreach
How many beds will be needed?	Days to weeks	Plan capacity

Multimodal GenAI moves from description to hypothesis

1 Describe visible image features

Possible today; errors remain.

2 Draft preliminary radiology findings

Possible with a specialized model and radiologist review.

3 Suggest a diagnostic hypothesis

Possible, but it demands clinical validation.

4 Diagnose or choose treatment autonomously

Outside this module and generic Copilot use.

Same image, different question

CNN QUESTION

Which review queue?

Output
Ranking score + route

Measured with
Confusion matrix



SAME 128 × 128 IMAGE

MULTIMODAL LLM QUESTION

What might be visible?

Output
Generated language

Evaluated here as
An unverified attempt

Separate experiment

The LLM receives no score, cutoff, route, or label.

A convincing answer can still be wrong

QUESTION	EVIDENCE HERE	CONCLUSION ALLOWED
Binary-class agreement?	Dataset label	Compare only
Detailed findings correct?	No report	Not verifiable
Claimed location correct?	No box or mask	Not verifiable
Ready for clinical use?	No validation	No

Generated does not mean validated

Fluent can still be incomplete, unsupported, or wrong.

Three real cases show three different lessons

IDx-DR · narrow autonomy

FDA-authorized screening with a specified population, protocol, and output.

[FDA review · DEN180001 ↗](#)

WHO TB CAD · calibrate locally

Conditional screening and triage use; calibrate by setting.

CLICK TO OPEN PRIMARY SOURCES

[WHO TB screening guideline ↗](#)

Epic sepsis · adoption is not proof

AUROC 0.63; 18% alerted; 67% of sepsis cases not identified.

[Wong et al. · PubMed ↗](#)

Sources: FDA; WHO TB guideline (2021); Wong et al. (2021)

SECTION III • 15 min

One image-classification case, end to end

A public pediatric chest-X-ray dataset becomes an educational queue-prioritization system with clinician review preserved.

The case: prioritize review, do not diagnose

Problem

Some pediatric chest X-rays may benefit from earlier review.

Model output

A ranking score from a 128×128 grayscale image.

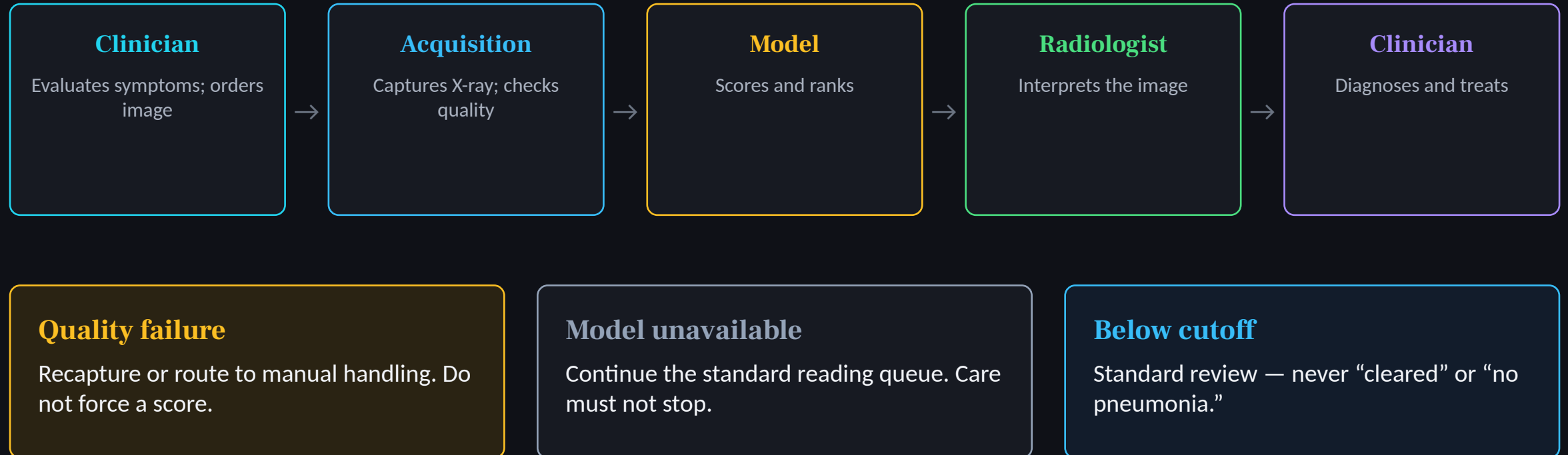
Workflow action

Priority or standard review; every study is interpreted.

The model sees one resized X-ray—not the whole patient

It does not receive symptoms, vitals, labs, history, or treatment context. Its output is review priority, not diagnosis.

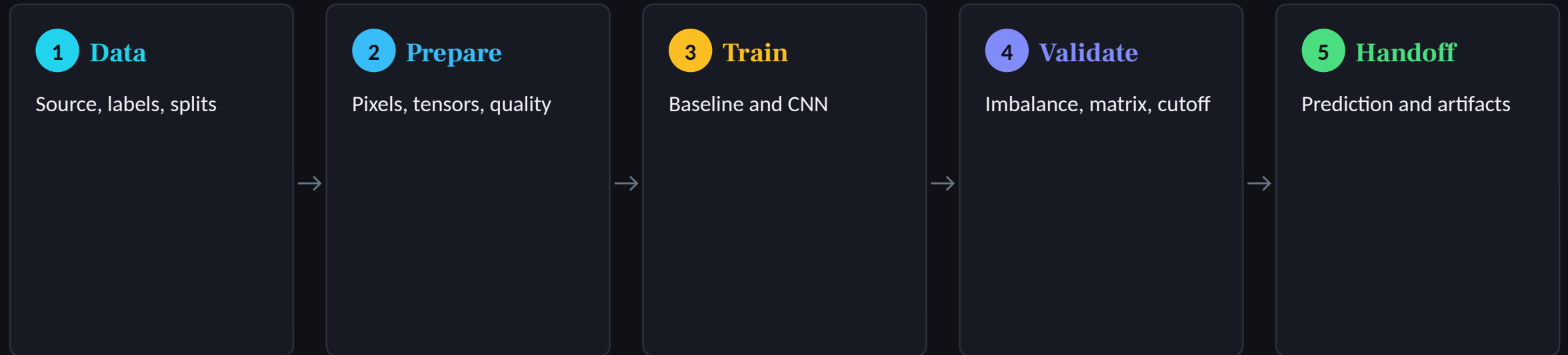
The complete workflow keeps two human decisions



Open the notebook

NOTEBOOK · X-ray pixels to review queue

notebooks/pneumonia-screening/01_pneumonia_build.ipynb



Optional Stage 6

Same image → new model → critique output.

After the notebook: each error changes someone's work

False negative

A pneumonia-labeled image remains in the standard queue.
Potential consequence: delayed interpretation and delayed care.

False positive

A normal-labeled image receives priority. Potential consequence: extra workload and delay displaced onto other studies.

Human review reduces risk; it does not remove it

Model errors still change queue order, staff attention, and time.

The app turns model results into a review workflow

Scoring workbench

Choose a held-out image, inspect quality, run the deployed model, and test controlled blur or exposure changes.

Priority-review queue

See score-ranked studies, queue volume, dataset outcomes, false positives, and false negatives.

Model card

Inspect population, data source, model, preprocessing, cutoff, metrics, limitations, and prohibited use.

Model estimate



Triage policy



Queue position



Radiologist

SECTION VI • 22 min

Human review needs a named role and clear authority

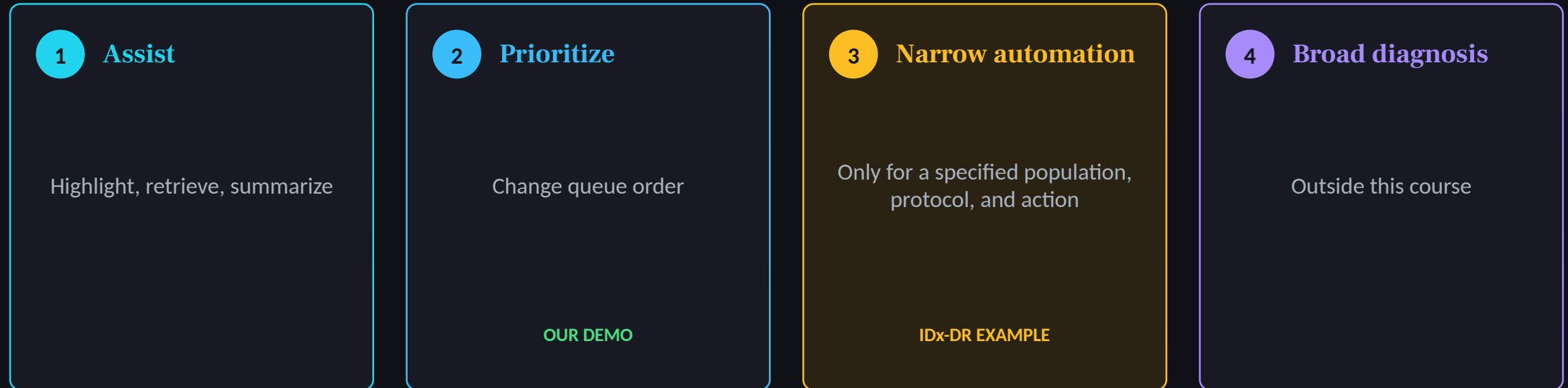
Name who reviews the output, what evidence they receive, what decision they own, and whether they have time to act.

Five roles, five different responsibilities

ROLE	OWNS	DOES NOT DELEGATE TO THE MODEL
Imaging staff	Acquisition and technical quality	Whether a failed image needs recapture or manual handling
Model	Score and queue ranking	Interpretation, diagnosis, or treatment
Radiologist	Image interpretation	Professional judgment about findings and uncertainty
Treating clinician	Diagnosis and treatment	Responsibility for the whole patient
Governance team	Validation and monitoring	Response to drift, incidents, and unsafe performance

DO NOT USE ONE RULE FOR EVERY SYSTEM

Medical AI can have different levels of autonomy



Source: FDA DEN180001; Abràmoff et al. (2018)

Before use, check data rights, evidence, and monitoring

1 · Data rights

De-identified does not mean licensed.
Check authority, license, and security.

2 · Intended use

Public data does not prove clinical validity.
Match population, setting, protocol, and output.

3 · Monitoring

Good launch results do not guarantee lasting safety. Track quality, errors, workload, overrides, and drift.

Sources: HHS OCR; FDA medical-device and CDS guidance

Key takeaways

1 **AI lives inside a workflow**

Map intake, diagnostics, operations, monitoring.

2 **An image is not a patient**

Pixels omit symptoms, history, labs, alternatives.

3 **Score, policy, action, outcome differ**

Scores prioritize; radiologists interpret; clinicians decide.

4 **Human oversight means authority**

People need time, evidence, and power.

5 **Validation leads to monitoring**

Watch model, workflow, outcomes, and shift.